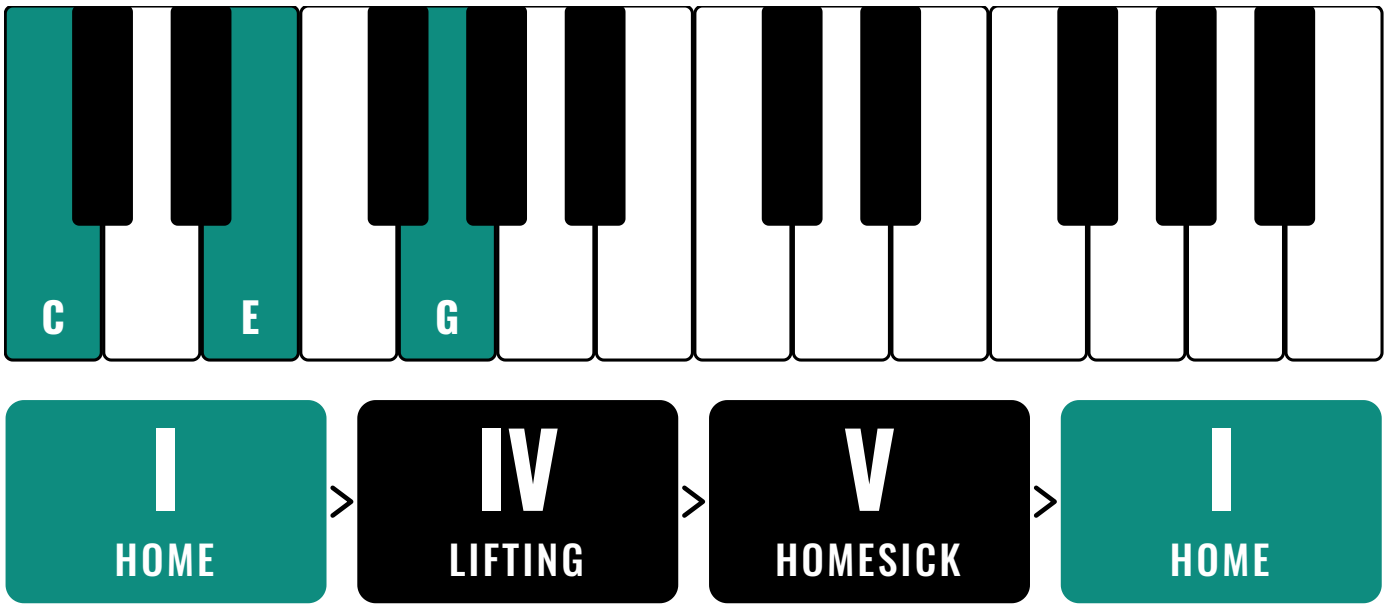
HUDSON BEAT



CHORD PROGRESSIONS IN LOGIC PRO

CHORDS BY EAR

DMP1 · Unit 3 · hear I–IV–V–I, then name it · C major + A minor

GRADES 9–12 · 5 DAYS · 35 PAGES + 21-SLIDE DECK

CHORDS BY EAR

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A complete 5-day chords unit for high school music technology, taught ear-first. On Day 1 students only listen and play: seven chords, two four-chord loops and a real song, described in their own words. Day 2 gives them the keyboard and the C major and A minor scales. Days 3 and 4 build chords from formulas and name them with Roman numerals (the KCSRN worksheet and a Reference Card). Then they build four chords over a bassline in Logic Pro and present on Day 5.

This is Unit 3 of my Digital Music Production 1 course. All students work in C major or A minor, white keys only. No music reading is needed. Unit 4, Bass & Melody Writing, is the next unit and a separate resource.

QUICK START ·

1. Read the overview and pacing (pages 3–6) and the teacher notes (pages 10–13).
2. Make the Part 3 Logic file: a 4-measure bassline (10 minutes, recipe on page 4).
3. Cue up Twist and Shout and Let It Be (The Beatles) on YouTube or Spotify.
4. Practice the opening activity once (page 10).
5. Print the student pages (pages 14–31): the packet, KCSRN and cards as one set per student.

THE INTRO DECK ·

Every slide has a day: Day 1 (slides 1–8), Day 2 (slides 9–10), Day 3 (slides 11–15), Day 4 (slides 16–19) and Day 5 (slides 20–21). The day plans name the slide for each block. Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint or Keynote shows a different font, install the two fonts or present the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–13	How to use, overview, setup + the Logic file recipe, Logic reference, pacing, day plans, teacher notes
Student pages	pages 14–31	Classroom posts, Chord Feel Log, Scary Stuff packet (Parts 1–3), KCSRN, Reference Cards, reference sheet, project sheet, vocabulary, checklist, rubric
Answer keys + example	pages 32–34	Packet and KCSRN answers, Feel Log sample words, a worked example in both keys
Intro deck (separate file)	deck	21 slides, .pptx + PDF, speaker notes on every slide

ABOUT THE SONGS

Songs are named by title and artist only. Play them from your own YouTube or Spotify access; no audio, video or lyrics are included.

MAKE IT FIT YOUR CLASS ·

- **File names:** the project sheet keeps my naming rule, "Your Name — Building Blocks Part 3" (The Building Blocks is my name for this theory sequence). Change it to suit your class.
- **More accessible / more advanced:** two variations on page 13.
- **No MIDI keyboard?** Musical Typing (Command-K) or the Pencil tool works for every step.
- **Short on time?** Run Days 1 and 2 as one day only if your students already know the keyboard; never skip the ear day.

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CHORDS BY EAR

Unit overview

OVERVIEW

DMP1 · Unit 3

WHAT STUDENTS MAKE ·

A Chord Feel Log in their own words, the Scary Stuff packet (keyboard, scales, chords, Roman numerals), the KCSRN worksheet and a Reference Card for their key. Then a short Logic project: four chords built over a provided bassline, one per measure, each labeled with its Roman numeral, plus a chord function table that says what each chord feels like.

WHAT THIS UNIT IS REALLY ABOUT ·

What a chord sounds like and feels like. The I chord feels like home. The V chord feels like homesick. The vi chord feels like something tender or melancholy. Major sounds bright; minor sounds dark. Chords are just notes from the scale stacked together. Roman numerals are names for feelings, not abstract theory.

Feel it before you name it: students play I–IV–V–I and describe how it feels before they know what those symbols mean. The KCSRN worksheet then arrives as "now name what you already heard," not as a cold fill-in exercise.

MIXING CONCEPT: BALANCE ·

When you layer parts, the background has to step back. When chords and melody are the same volume, the melody disappears. Turning the chords down is not a mixing trick; it is a musical decision about what the listener should focus on.

GOALS ·

- Describe what a chord feels like in plain language before naming it
- Name every white key and tell a half step from a whole step
- Build the C major and A natural minor scales from their formulas
- Build major, minor and diminished chords with R-4-3, R-3-4 and R-3-3
- Use Roman numerals to describe chord function in C major or A minor
- Build and label four chords over a bassline in Logic

AT A GLANCE ·

Grades	9–12 (no music reading needed)
Length	5 periods of about 53 minutes: listen, keyboard and scales, chords, Roman numerals, present
Keys	C major or A minor, white keys only
Software	Logic Pro: software instrument tracks, MIDI regions, the piano roll, region names, volume, export
Product	Packet, KCSRN and Reference Card; four labeled chords over a bassline; chord function table; MP3 + compressed project
Assessment	3-category rubric, /15 (page 31)

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): analyzing how musical elements are used (MU:Re7.2), developing musical ideas (MU:Cr2.1) and presenting work (MU:Pr6.1).

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CHORDS BY EAR

Setup and requirements

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** and **headphones** for each student; a MIDI keyboard (an MPK Mini or similar) if you have one.
- **A piano sound on the class speakers** for the Day 1 opening and the Day 4 Roman numeral demo.
- **Projector** for the deck; **two songs** from your own YouTube or Spotify access: Twist and Shout and Let It Be (The Beatles).
- **Printed student pages**, one set each: Chord Feel Log, packet, KCSRN, Reference Cards (C major, A minor, blank), reference sheet, project sheet.
- **The Part 3 Logic file** with a bassline (recipe on the right).
- **A place to submit** an MP3 and a .zip.

BEFORE DAY 1 ·

- ☐ Open Logic with a piano sound and test the class speakers.
- ☐ Practice the opening activity (page 10) and cue up the songs.
- ☐ Check that students can play chords: MPK Mini, or Musical Typing (Command-K).
- ☐ Make and test the Part 3 Logic file; decide C major, A minor or both.
- ☐ Print the student pages; set up the submission spot.

MAKE THE PART 3 LOGIC FILE (10 MINUTES)

I give students a Logic file with the bassline already written. Logic projects are not included here, so build your own once. My recommended setup:

- Same key as the scales work (C major or A minor)
- Bassline clearly following chord roots so students can infer the progression from the bass
- Arrangement markers with chord labels visible
- No chord track — leave that blank for students to build

Recipe:

1. New project, one Software Instrument track with a bass sound (Library: Bass). Tempo between 80 and 100 BPM is comfortable.
2. Draw a 4-measure MIDI region. In the piano roll, add one whole note per measure, the root of each chord. C major example: C – A – F – G (low octave). A minor example: A – F – C – G.
3. Turn on Cycle over the 4 measures so it loops.
4. Optional, as in my file: arrangement markers with chord labels visible.
5. Leave the chord track empty: students build it. Save it as a template or share the project with your class.

The examples use the same four chords in both keys: I–vi–IV–V in C major and i–VI–III–VII in A minor (worked example on page 34). Any four chords from the Reference Card work.

WHAT COMES NEXT

Unit 4, Bass & Melody Writing, picks up here: the rule behind every bassline, and a short melody over the chords students just built.

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CHORDS BY EAR

Logic quick reference

SETUP

Teacher

Everything students need in Logic for this unit, in one table. On Day 3 demonstrate one chord (slide 15), then let students build the rest.

New instrument track	Track › New Software Instrument Track (Option-Command-S). Open the Library (Y) and pick a piano or electric piano for chords, a bass sound for the bassline.
Play without a keyboard	Window › Show Musical Typing (Command-K) turns the computer keys into a piano. Z and X move the octave.
Empty MIDI region	With the Pencil tool, click in the track's lane, or Control-click the lane and choose Create MIDI Region. Drag its right edge to make it one measure long.
Piano roll	Double-click the region, or press P. With the Pencil tool, click to add a note; drag its right edge to set the length.
Stack a chord	All three notes start at the same position and last the same length. Select the notes and press Q to quantize them to the grid.
Copy a chord	Option-drag a region to copy it; then drag its notes up or down in the piano roll to make the next chord.
Name a region	Select it and type the Roman numeral in the Region inspector's name field (or press Shift-N).
Loop it	Press C for Cycle and drag the cycle area over the four measures.
Balance	Open the Mixer (X). Pull the chord track's fader down until the bass is clear. Solo (S) a track to hear it alone.
Export	File › Bounce › Project or Section (Command-B), MP3. In Finder, Control-click the .logicx project and choose Compress.

CHORD CHECK IN THE PIANO ROLL

Three notes stacked in one column, same start, same length. Count the rows between them: 4 then 3 half steps is major, 3 then 4 is minor, 3 then 3 is diminished. Every grey row (a black key) counts.

VERSION NOTE

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc., named only to describe what the unit needs. This resource is not affiliated with Apple. MPK Mini is named only as an example keyboard.

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CHORDS BY EAR

Pacing guide · 5 days

PACING

Teacher

Five class periods of about 53 minutes: two learning days (ear first, then the keyboard and scales), two project days (chords, then Roman numerals) and a finishing and presenting day. Each day's full plan is on pages 7–9.

DAY	FOCUS AND GOAL	MINUTE BLOCKS	DECK
DAY 1 Learning	**Chords by ear: just listen** Every student has played the seven white-key chords and described each one in plain words, and heard the four-chord loop come home. No Roman numerals yet.	0–10 · Opening: one note, then chords 10–18 · Listen: a real song 18–45 · Play and describe 45–53 · Share-out	Intro deck slides 1–8
DAY 2 Learning	**The keyboard and scales** Every student can name all 7 white keys, tell a half step from a whole step, and build the C major and A natural minor scales.	0–5 · Bright or dark? 5–20 · The keyboard 20–30 · Logic: name every key 30–50 · Scales 50–53 · Exit question	Intro deck slides 9–10
DAY 3 Project day	**Build chords from formulas** Chord construction exercises complete. Students can build any major or minor chord from any root and enter it in the piano roll.	0–5 · What notes are those? 5–12 · The project 12–35 · Chord construction 35–50 · Logic: chords in the piano roll 50–53 · Save + check	Intro deck slides 11–15
DAY 4 Project day	**Roman numerals + Reference Card** Reference Card complete. Chords built over the bassline in Logic and labeled with their Roman numerals.	0–15 · Name what you already heard 15–40 · Packet, KCSRN + Reference Card 40–53 · Chords over the bassline	Intro deck slides 16–19
DAY 5 Finish + present	**Balance, finish, present** Every project submitted and presented; every student can point to their I chord and say why it feels like home.	0–8 · Mixing moment: balance 8–30 · Finish + submit 30–50 · Presentations 50–53 · Close	Intro deck slides 20–21

STUDENT PAGES BY DAY ·

Day 1	Day 1 post (p. 14), Chord Feel Log (p. 15)
Day 2	Day 2 post (p. 16), packet Parts 1–2 (pp. 18–19), reference sheet (p. 26)
Day 3	Project sheet (pp. 27–28), chords post (p. 17, step 1), packet Part 3 chords (p. 20)
Day 4	Chords post steps 2–5, Roman numerals (p. 21), KCSRN (p. 22), Reference Cards (pp. 23–25)
Day 5	Chord function table (p. 27), checklist (p. 30), rubric (p. 31)

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CHORDS BY EAR

Day-by-day plans · Days 1 and 2

DAYS 1–2

Learning days

DAY 1 · CHORDS BY EAR: JUST LISTEN

Learning · Intro deck slides 1–8

Goal: Every student has played the seven white-key chords and described each one in plain words, and heard the four-chord loop come home. No Roman numerals yet.

0–10 **OPENING: ONE NOTE, THEN CHORDS** [slides 1–4] · Slides 1–4. Teacher-led, no worksheets yet (steps on page 10). Play a single C, then C major, then A minor, then the four chords C–F–G–C slowly. Ask what each one feels like and what happened at the end.

10–18 **LISTEN: A REAL SONG** [slide 5] · Slide 5: Twist and Shout by The Beatles, the first minute. How many different chords? Which one feels like home? When does it want to go back?

18–45 **PLAY AND DESCRIBE** [slides 6–7] · Slides 6–7. Students load a piano sound in Logic and play the seven chords on the Chord Feel Log (page 15), writing what each one feels like in their own words. Then they loop the two four-chord progressions and compare.

45–53 **SHARE-OUT** [slide 8] · Slide 8. Collect the class's words on the board and keep them: on Day 4 the Roman numerals give them shorter names.

DAY 2 · THE KEYBOARD AND SCALES

Learning · Intro deck slides 9–10

Goal: Every student can name all 7 white keys, tell a half step from a whole step, and build the C major and A natural minor scales.

0–5 **BRIGHT OR DARK?** · Play the C major scale, then the A natural minor scale. Ask: which sounds brighter? Which sounds darker? Do not explain yet.

5–20 **THE KEYBOARD** [slide 9] · Slide 9. Students complete Part 1 of the packet (page 18): label the white keys, then the half step / whole step exercises. Find C first.

20–30 **LOGIC: NAME EVERY KEY** · Students load a piano sound and play every white key from C to C, naming each one aloud. Then C and D, then C and C#: can you hear the difference between a whole step and a half step?

30–50 **SCALES** [slide 10] · Slide 10. Students complete Part 2 of the packet (page 19): C major, A natural minor and C major pentatonic, playing each scale as they fill it in.

50–53 **EXIT QUESTION** · Which two notes from C major are missing from C major pentatonic? Collect answers at the door.

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CHORDS BY EAR

Day-by-day plans · Days 3 and 4

DAYS 3–4

Project days

DAY 3 · BUILD CHORDS FROM FORMULAS

Project day · Intro deck slides 11–15

Goal: Chord construction exercises complete. Students can build any major or minor chord from any root and enter it in the piano roll.

0–5 **WHAT NOTES ARE THOSE?** · Play C major and A minor again. Ask: now that you know notes, can you figure out what notes those are?

5–12 **THE PROJECT** [slides 11–13] · Slides 11–13: the project, what to turn in and the rubric. Hand out the project sheet (pages 27–28).

12–35 **CHORD CONSTRUCTION** [slide 14] · Slide 14. Students work through Part 3 of the packet (page 20): major, minor and diminished chords. Ask every student to count their first chord aloud before they write it.

35–50 **LOGIC: CHORDS IN THE PIANO ROLL** [slide 15] · Slide 15: demonstrate one chord in the piano roll (page 5). Students build the chords from their packet work as MIDI regions on a new track and play each one to confirm it sounds correct.

50–53 **SAVE + CHECK** · Students save. Exit check: spell one major and one minor chord out loud.

DAY 4 · ROMAN NUMERALS + REFERENCE CARD

Project day · Intro deck slides 16–19

Goal: Reference Card complete. Chords built over the bassline in Logic and labeled with their Roman numerals.

0–15 **NAME WHAT YOU ALREADY HEARD** [slides 16–17] · Slides 16–17. Play I–IV–V–I: home, away, very away, home. Play i–VI–III–VII: how is that different? Name the feelings before naming the numerals, then match the class's Day 1 words to the numerals. Let It Be by The Beatles for the vi chord.

15–40 **PACKET, KCSRN + REFERENCE CARD** [slide 18] · Slide 18. Students complete the Roman numeral section of the packet (page 21), the KCSRN worksheet for their key (page 22) and their Reference Card (page 25). Circulate and check that chord spellings are correct.

40–53 **CHORDS OVER THE BASSLINE** [slide 19] · Slide 19: the worked example. Students open the Logic file, listen to the bassline twice, build the 4 chords (one per measure) and label each region with its Roman numeral.

SPELL-CHECK AT A GLANCE (FOR CIRCULATING)

C major	**I** C–E–G	**ii** D–F–A	**iii** E–G–B	**IV** F–A–C	**V** G–B–D	**vi** A–C–E	**vii°** B–D–F
A minor	**i** A–C–E	**ii°** B–D–F	**III** C–E–G	**iv** D–F–A	**v** E–G–B	**VI** F–A–C	**VII** G–B–D

Same seven chords in both keys, white keys only; only the Roman numerals move. Full answers on pages 32–33.

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CHORDS BY EAR

DAY 5

Finish + present

DAY 5 · BALANCE, FINISH, PRESENT

Finish + present · Intro deck slides 20–21

Goal: Every project submitted and presented; every student can point to their I chord and say why it feels like home.

0–8 **MIXING MOMENT: BALANCE** [slide 20] · Slide 20. Solo the chords, then the bass, then play both: which one should the listener follow? Turn the chords down until the bass is clear.

8–30 **FINISH + SUBMIT** · Students complete the chord function table on the project sheet, finalize the Logic project, export and submit. Circulate and ask every student: point to your I chord. Play it. Does it feel like home?

30–50 **PRESENTATIONS** · Each student plays their four chords and explains them using Roman numeral language: which chord is home, which one wants to go back, and what each one feels like.

50–53 **CLOSE** [slide 21] · Slide 21: point to your I chord. Does it feel like home? That is the test.

CIRCULATING PROMPTS (DAYS 3–5)

- ‘Count your first chord out loud for me, starting from the root.’
- ‘Do all three notes start in the same place? Does it look like a chord or a melody?’
- ‘Point to your I chord. Play it. Does it feel like home?’
- ‘Which chord wants to go back home? What is its Roman numeral?’

PRESENTATION QUESTIONS (DAY 5)

- What are your four chords, as Roman numerals and as chord names?
- Which one feels like home? Which one wants to resolve?
- What would you change if you could swap one chord?

The rubric’s top level asks students to explain their chord choices using Roman numeral language: listen for “the V wants to go back to the I,” not just “it sounds good.”

NOTES FOR NEXT YEAR ·

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FEEL IT BEFORE YOU NAME IT ·

This is the single most important fix in my course sequence: theory used to arrive as a cold worksheet. Taught ear-first, every later project has real harmonic understanding behind it. The most common failure mode is rushing to the worksheet. Spend the full first day just playing chords and describing feelings in plain language. Those descriptions are music theory; the Roman numerals just give them shorter names.

DAY 1: THE OPENING ACTIVITY (FIRST 10 MINUTES) ·

Ten minutes of playing notes and chords before any theory framing plants an emotional reference point that the packet then labels. Do not skip it, even if it feels like lost time.

1. Open Logic. Load a piano sound. Make sure students can hear.
2. Play a single C. Ask: what does that feel like? (Empty. Waiting.) Good.
3. Play C major (C-E-G). Ask: what changed? Let them answer freely. Happy, full, finished.
4. Play A minor (A-C-E). Ask: what is different? Dark, heavy, sad.
5. Play I-IV-V-I slowly. Ask: what happened at the end? Most will say it resolved. Why?
6. Tell them: the Scary Stuff packet gives names to everything you just felt. That is all music theory is.

HOW THIS KIT DIFFERS FROM MY CLASSROOM VERSION

- My Building Blocks material runs Parts 1–3 in 6 days; this kit follows the 5 days of my course plan: the ear day, then the keyboard and scales in one day, then the three Part 3 days (presentations on Day 5).
- The Chord Feel Log, the Day 1 post and the Logic file recipe are new: my plan calls for an ear-first activity before the KCSRN worksheet. Part 2's melody project and Part 4 (basslines) move to Unit 4, so on the project sheet the overview's first sentence ("You have written a melody…") now reads "You have learned the keyboard and the notes of your scale", and the tip "You will need these labels for Part 4" now says "when you present your project".
- On the A minor Reference Card the closing reminder uses the minor-key numerals (i, v and VII, VI) instead of my C major line (I, V, vi), which names chords that are not on that card. Part 3 of the packet now tells students that the D major, B minor and D diminished boxes use black keys and are formula practice only.
- My LMS is written as "your class page". The balance mini-lesson on Day 5 applies my mixing concept for this unit to chords and bass, since melody comes in Unit 4.

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ROMAN NUMERALS: THE KEY FRAMING ·

Part 3 has two distinct moments. The first is mechanical: students learn to build chords from formulas and enter them correctly in the piano roll. The second is conceptual: Roman numerals are introduced as a way to describe chord function rather than just chord names. The second moment is more important and should not be rushed.

A chord's letter name tells you what notes it contains. Its Roman numeral tells you how it functions: what emotional role it plays in the key. The I chord always feels like home regardless of whether the key is C major or G major or Bb major. That principle transfers to every key students will ever work in.

On Day 4, play I–IV–V–I together: home, away, very away, home. Play i–VI–III–VII: how is that different? Name the feelings before naming the numerals. Then bring back the class's words from Day 1: the KCSRN worksheet is "now name what you already heard."

THE REFERENCE CARD IS A LIVING DOCUMENT ·

Students will use it again in the next unit, in the EDM project and beyond. Make clear that filling it in carefully now is an investment. The completed C major and A minor cards (pages 23–24) are for checking; the blank card (page 25) is the one students fill in.

ABOUT THE LOGIC FILE ·

The Part 3 file contains a bassline already written; students add only the chord layer. The bass follows the chord roots so students can infer the progression from the bass. Recipe on page 4.

MIXING CONCEPT: BALANCE (DAY 5) ·

Solo the chords, solo the bass, then play both. Ask which one the listener should follow. Students pull the chord fader down until the bass is clear. It is not graded; it is the habit the next unit needs when the melody arrives.

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COMMON STUDENT STRUGGLES ·

Rushing to the worksheet: The most common failure mode in this unit. Spend the full first day just playing chords and describing feelings in plain language. ‘That one sounds like something good is about to happen.’ ‘That one sounds unfinished.’ Those descriptions are music theory. The Roman numerals just give them shorter names.

Finding C on the MPK Mini: The MPK Mini keys are small and not labeled. Show students the pattern: C always sits just to the left of the group of 2 black keys. Have every student find C before they do anything else.

Counting half steps from the root incorrectly: The single most common chord construction error. Students count the root as step 1. Demonstrate explicitly: for C major (R-4-3), start from C# as step 1 — C#(1), D(2), D#(3), E(4). Then count 3 from E: F(1), F#(2), G(3). Walk around and ask every student to verbally count their first chord before they enter it in Logic.

Chords entered as melody notes: Students sometimes enter chord notes one at a time horizontally instead of stacked vertically. Show the piano roll with a correct chord: all three notes start at the same position and have the same length. Any chord that looks like a melody line is wrong.

Roman numerals feeling abstract: Students who built their chords mechanically and did not play them will struggle to connect Roman numerals to feelings. Return to the opening activity: play I and V on the class speaker and ask what the difference feels like. Then explain: I is labeled I because it is the home chord, built on the first note of the scale. That is all the Roman numeral tells you.

Rushing through the packet: The half/whole step exercises are not optional. They build the mechanical understanding that makes chord construction possible. Check completion before students open Logic.

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SUGGESTED VARIATIONS ·

More accessible: Pre-fill the chord spellings in the Scary Stuff packet for the key students are working in. Ask them to verify each chord in Logic rather than calculate it from scratch.

More advanced: Ask students to add a second chord layer with a different instrument sound and slightly varied rhythm. Introduce the concept of harmonic rhythm (how often the chords change) and ask students to experiment with half-measure chord changes.

GRADING TIPS ·

- Three categories, each scored 2–5, for a total out of 15 (page 31).
- Chord Construction: open the piano roll and count. The answer keys (pages 32–34) list every chord in both keys.
- Harmonic Understanding: the chord function table. ‘Sounds good’ is a 3; ‘wants to go back home’ is the kind of specific answer the 5 asks for.
- The quickest evidence: point to your I chord. Play it. Does it feel like home? That is the test.

RESOURCES ·

- The intro deck (.pptx and PDF), included.
- musictheory.net, lesson 44 (free), as extra reading for the KCSRN worksheet.
- The two songs, from your own YouTube or Spotify access.

OBSERVATION NOTES ·

What worked, what to adjust, and anything worth remembering for next year.

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CHORDS BY EAR · DAY 1

Digital Music Production 1 · Unit 3

Today is a listening day. You are going to play chords and describe what they feel like before anyone tells you what they are called.

Complete the Chord Feel Log before you leave today.

1. **LISTEN** · Your teacher will play one note, then chords, in Logic. Describe what you hear in plain language.
   No wrong answers. Use any words that come to mind.

2. **LISTEN TO A REAL SONG** · Your teacher will play Twist and Shout by The Beatles. How many different chords do you hear? Which one feels like home?
   You do not need any names yet. Just notice.

3. **PLAY THE SEVEN CHORDS** · Open Logic, load a piano sound, and play each chord on the Chord Feel Log (page 15). Write what it feels like in your own words.
   Play it more than once. Say more than 'good' or 'bad': sad how? Waiting for what?

4. **LOOP THE TWO PROGRESSIONS** · Play Loop 1 and Loop 2 from the log on repeat. Where does each one feel at home?
   Same white keys, a different home.

5. **SHARE YOUR WORDS** · Bring your best words to the share-out. We will keep them on the board.
   On Day 4 these words get shorter names. You are already doing music theory.

TODAY'S ONE IDEA

Three or more notes played at the same time make a **chord**. Today you only need your ears.

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CHORDS BY EAR

Chord Feel Log · play it, then describe it

STUDENT

Day 1 · by ear

Name _____ Date _____

Play each chord (all three notes at once) on a piano sound. Play it more than once. Then write what it feels like in your own words and circle bright or dark. No names needed today.

#	PLAY THESE NOTES	WHAT DOES IT FEEL LIKE? (YOUR WORDS)	CIRCLE ONE
1	**C–E–G**		bright · dark
2	**F–A–C**		bright · dark
3	**G–B–D**		bright · dark
4	**A–C–E**		bright · dark
5	**D–F–A**		bright · dark
6	**E–G–B**		bright · dark
7	**B–D–F**		bright · dark

LOOP 1 · PLAY IT ON REPEAT

C–E–G › F–A–C › G–B–D › C–E–G

What happens at the end? _____

Home is: _____

LOOP 2 · PLAY IT ON REPEAT

A–C–E › F–A–C › C–E–G › G–B–D

How is it different? _____

Home is: _____

Which chord felt most like home? _____

Which one wanted to go somewhere? _____

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THE KEYBOARD AND SCALES · DAY 2

Digital Music Production 1 · Unit 3

Today we start at the beginning — the notes on a keyboard. Before you can write music, you need to know what keys you are pressing.

Complete the Scary Stuff packet Part 1 before opening Logic.

1. **LISTEN** · Your teacher will play two scales. Which sounds brighter? Which sounds darker?
   Do not worry about names yet.

2. **PART 1 — KEYBOARD DIAGRAM** · Label all 7 white keys on the keyboard diagram in the packet (page 18).
   Find C first — it sits just to the left of every group of 2 black keys.

3. **PART 1 — HALF/WHOLE STEPS** · Complete the half step / whole step exercises. The first two are done for you as examples.
   Play each interval on your MPK Mini as you label it.

4. **LOGIC ACTIVITY** · Open Logic, load a piano sound, and play every white key from C to C, naming each one as you press it.
   Then play C and D. Then C and C#. Can you hear the difference between a whole step and a half step?

5. **PART 2 — SCALES** · Build the C Major, A Natural Minor and C Major Pentatonic scales in the packet (page 19).
   Play each scale on your MPK Mini as you complete it.

THE WHITE KEYS IN ORDER

C – D – E – F – G – A – B – (back to C)

E to F and B to C are the only half steps between white keys — no black key between them.

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CHORDS & ROMAN NUMERALS · DAYS 3–4

Digital Music Production 1 · Unit 3

Today you are building chords in Logic — and learning the system that explains why certain chords feel the way they do.

Complete Part 3 of the packet before building anything in Logic.

1. **DAY 3 · PART 3 — CHORD CONSTRUCTION** · Complete the major, minor, and diminished chord exercises in the packet. C Major and A Minor are pre-filled as examples.
   Play each chord in Logic as you complete it on paper. Does it sound like what the formula predicts?

2. **DAY 4 · PART 3 — ROMAN NUMERALS** · Read the Roman numeral section and complete the chord feel table.
   The I chord always feels like home. The V chord always wants to resolve. These feelings are the same in every key.

3. **DAY 4 · COMPLETE YOUR REFERENCE CARD** · Fill in your Reference Card (C Major or A Minor) using your packet work and the reference sheets.
   Keep this card. You will use it for the rest of the year.

4. **DAY 4 · OPEN THE LOGIC FILE** · Download today's Logic file. A bassline is already written. Listen to it twice before adding anything.
   What are the chords? What Roman numerals would you use to label them?

5. **DAY 4 · BUILD THE CHORDS** · Add a new track and build the chords for the progression using the piano roll. One chord per measure. Each chord should be 3 notes stacked vertically.
   Use your Reference Card to check your chord spellings.

THE CHORD FORMULAS

Major: R - 4 - 3 — bright and stable

Minor: R - 3 - 4 — darker and more emotional

Diminished: R - 3 - 3 — tense and unstable

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CHORDS BY EAR

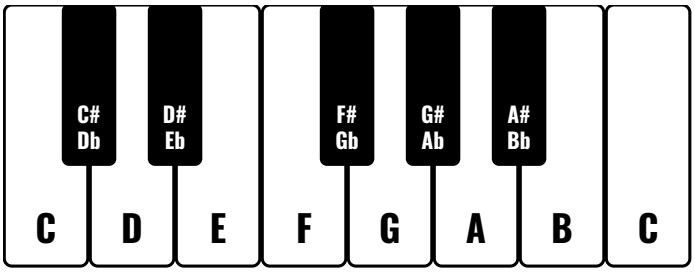
The Scary Stuff · Part 1: The Keyboard

PACKET

Part 1 · Day 2

Name _____ Date _____

The keyboard repeats the same pattern of 7 white keys and 5 black keys over and over. Learn one octave and you know all of them.



Find C on your MPK Mini. It sits just to the left of every group of 2 black keys. Play C, then continue upward naming each white key: C D E F G A B. Then it starts again at C.

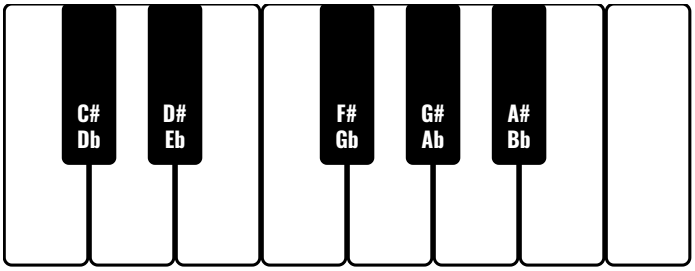
HALF STEP (H)

Moving one key in either direction — including black keys. The smallest possible interval. *E to F is a half step. B to C is a half step. These are the only two white-key pairs with no black key between them.*

WHOLE STEP (W)

Two half steps — skip one key. Most white key pairs are whole steps. *C to D is a whole step. D to E is a whole step. G to A is a whole step.*

NOW TRY IT. LABEL THE WHITE KEYS WITHOUT LOOKING ABOVE ·



HALF STEP (H) OR WHOLE STEP (W)? THE FIRST TWO ARE DONE FOR YOU ·

C to D	**W**	**C# to D**	**H**
B to C		**E to F**	
D to E		**A to B**	
F to G		**G# to A**	

When you finish: open Logic, load any piano sound, and play every white key from C to C naming each one out loud as you press it.

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CHORDS BY EAR

The Scary Stuff · Part 2: Scales

PACKET

Part 2 · Day 2

Name _____ Date _____

A scale is an ordered set of notes built using a formula of whole steps (W) and half steps (H). The formula tells you exactly which notes to use. Play each scale on your MPK Mini as you complete it.

C MAJOR SCALE · W - W - H - W - W - W - H ·

Starting on C, apply the formula. The first two notes are done for you.

C	D						(=C)
	W	W	H	W	W	W	H

Play it. Describe what it sounds like in one word: _____

A NATURAL MINOR SCALE · W - H - W - W - H - W - W ·

Starting on A, apply the formula. Fill in all 8 notes.

A							(=A)
	W	H	W	W	H	W	W

Play it. How does it feel different from C Major? _____

ABOUT THE PENTATONIC SCALE

A pentatonic scale removes 2 notes from the major or minor scale, leaving only the ones that sound most agreeable together. WH = one whole step + one half step combined (3 half steps).

C MAJOR PENTATONIC · W - W - WH - W - WH (5 NOTES) ·

C					(=C)
	W	W	WH	W	WH

Which two notes from C Major are missing from C Major Pentatonic? _____ _____

Keep going: the notes of these scales are the notes of your chords. Every chord in C major is built only from C D E F G A B.

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CHORDS BY EAR

PACKET

The Scary Stuff · Part 3: Chords

Part 3 · Day 3

Name _____ Date _____

A chord is three or more notes played at the same time. You build chords by stacking intervals on top of a root note. Play each chord in Logic as you complete it.

R = Root · 4 = 4 half steps up from the root · 3 = 3 half steps up from the previous note. *Do not count the note you start on: for C, C# is 1.*

MAJOR CHORDS · R-4-3 — SOUNDS BRIGHT AND STABLE ·

C MAJOR

C	E	G
Root	3rd	5th

F MAJOR

Root	3rd	5th

G MAJOR

Root	3rd	5th

D MAJOR

Root	3rd	5th

MINOR CHORDS · R-3-4 — SOUNDS DARKER AND MORE EMOTIONAL ·

A MINOR

A	C	E
Root	3rd	5th

D MINOR

Root	3rd	5th

E MINOR

Root	3rd	5th

B MINOR

Root	3rd	5th

DIMINISHED CHORDS · R-3-3 — SOUNDS TENSE AND UNSTABLE ·

B DIMINISHED

Root	3rd	5th

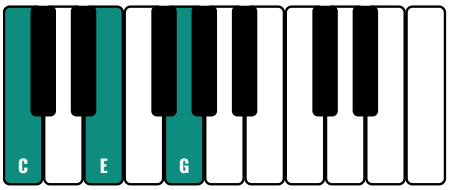
D DIMINISHED

Root	3rd	5th

Heads up: D major (F#), B minor (F#) and D diminished (Ab) use black keys. They are formula practice only. The chords in your own key, C major or A minor, are white keys only.

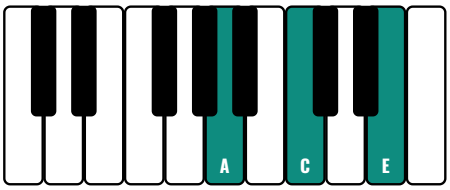
C MAJOR

R-4-3 · C - E - G



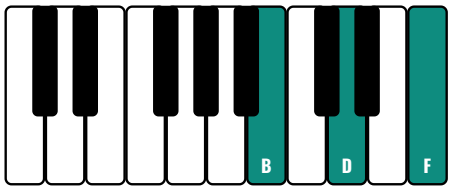
A MINOR

R-3-4 · A - C - E



B DIM.

R-3-3 · B - D - F



Teal keys: the examples, counted from the formula. Use the keyboard to count yours.

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CHORDS BY EAR

The Scary Stuff · Part 3: Roman Numerals

PACKET

Part 3 · Day 4

Name _____ Date _____

ROMAN NUMERALS — NAMES FOR CHORD FUNCTION ·

Every chord in a key has a Roman numeral based on its position in the scale. The Roman numeral tells you something the letter name cannot — it tells you how the chord functions. The I chord always feels like home. The V chord always feels like it needs to resolve. These feelings are the same in every key.

Uppercase = major chord. Lowercase = minor chord. ° = diminished. In the last column, write the word the class used for that feeling on Day 1.

RN	TYPE	CHORD	FEELS LIKE…	OUR DAY 1 WORD
I / i	Major/Minor	**Home chord**	*Home. Settled. Complete. The resting point of the key.*	
ii / ii°	Minor/Dim.	**Tension chord**	*Gentle tension. Moving somewhere.*	
IV / iv	Major/Minor	**Lifting chord**	*Hopeful. Away from home but not lost.*	
V / v	Major/Minor	**Dominant chord**	*Homesick. Wants to resolve back to I. Very strong pull.*	
vi / VI	Minor/Major	**Emotional chord**	*Tender, melancholy. The emotional heart of the key.*	

HEAR IT ·

Play I–IV–V–I in C major. Write the chord names: _____

Play i–VI–III–VII in A minor. How is it different? _____

Which chord pulls hardest back to home? _____

Now complete your Reference Card for your chosen key. Use the Scary Stuff Reference Sheet if you need help.

When you finish: open the Logic file your teacher provides. A bassline is already written. Build the chords on top of it using the formulas above.

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CHORDS BY EAR

KCSRN worksheet

KCSRN

Day 4

Name _____ Date _____

Keys, chords, scales and Roman numerals. Now name what you already heard. Fill in the column for your key (major = C major, minor = A minor). Extra reading: musictheory.net, lesson 44.

MAJOR KEY

Major Key: _____

Scale:

___ ___ ___ ___ ___ ___ ___ ___

CHORD NAMES

I _____

ii _____

iii _____

IV _____

V _____

vi _____

vii° _____

CHORD SPELLINGS

I: ___ ___ ___

ii: ___ ___ ___

iii: ___ ___ ___

IV: ___ ___ ___

V: ___ ___ ___

vi: ___ ___ ___

vii°: ___ ___ ___

REMINDERS!

Major Scale: W W H W W W H

Major chord: R 4 3 · Minor chord: R 3 4

Diminished (°) chord: R 3 3

MINOR KEY

Minor Key: _____

Scale (natural minor):

___ ___ ___ ___ ___ ___ ___ ___

CHORD NAMES

i _____

ii° _____

III _____

iv _____

v _____

VI _____

VII _____

CHORD SPELLINGS

i: ___ ___ ___

ii°: ___ ___ ___

III: ___ ___ ___

iv: ___ ___ ___

v: ___ ___ ___

VI: ___ ___ ___

VII: ___ ___ ___

REMINDERS!

Natural Minor Scale: W H W W H W W

Major chord: R 4 3 · Minor chord: R 3 4

Diminished (°) chord: R 3 3

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C MAJOR — REFERENCE CARD

The Building Blocks · Unit 3 · Keep this open while you work

SCALE ·

C Major — W - W - H - W - W - W - H

C	D	E	F	G	A	B	C	(8)

CHORDS IN THIS KEY ·

RN	TYPE	CHORD	NOTES	FEELS LIKE…
I	Major	**C Major**	C - E - G	*Home. Settled. Complete.*
ii	Minor	**D Minor**	D - F - A	*Gentle tension. Moving somewhere.*
iii	Minor	**E Minor**	E - G - B	*Wistful. In between.*
IV	Major	**F Major**	F - A - C	*Lifting. Hopeful. Away from home.*
V	Major	**G Major**	G - B - D	*Homesick. Wants to resolve to I.*
vi	Minor	**A Minor**	A - C - E	*Tender. Melancholy. Emotional.*
vii°	Dim.	**B Dim.**	B - D - F	*Tense. Unstable. Rarely used alone.*

Remember: Roman numerals describe chord function, not just chord names.

The I chord always feels like home. · The V chord always feels unresolved. · The vi chord always feels tender or melancholy.

These feelings are the same in every key — only the note names change.

No sharps or flats in this key: every chord is white keys only. A minor uses the same seven notes: it is the relative minor.

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A MINOR — REFERENCE CARD

The Building Blocks · Unit 3 · Keep this open while you work

SCALE ·

A Minor — W - H - W - W - H - W - W

A	B	C	D	E	F	G	A

CHORDS IN THIS KEY ·

RN	TYPE	CHORD	NOTES	FEELS LIKE…
i	Minor	**A Minor**	A - C - E	*Home. But darker and more unsettled than major.*
ii°	Dim.	**B Dim.**	B - D - F	*Tense. Rarely used alone.*
III	Major	**C Major**	C - E - G	*Relative major. Bright moment in a darker key.*
iv	Minor	**D Minor**	D - F - A	*Heavy. Emotional. Pulling back toward home.*
v	Minor	**E Minor**	E - G - B	*Less forceful tension than major V.*
VI	Major	**F Major**	F - A - C	*Warm and open. Common resolution point.*
VII	Major	**G Major**	G - B - D	*Lifting. Drives back toward i strongly.*

Remember: Roman numerals describe chord function, not just chord names.

The i chord always feels like home. · The v and VII chords pull back toward i. · The VI chord feels warm and open.

These feelings are the same in every key — only the note names change.

No sharps or flats in this key: every chord is white keys only. C major uses the same seven notes: it is the relative major.

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CHORDS BY EAR

Reference Card · blank

REFERENCE

Fill it in

Name _____ Date _____

MY KEY — REFERENCE CARD

The Building Blocks · Unit 3 · Keep this open while you work

My key: _____

SCALE ·

Key: _____ **Formula:** _____

ROOT							***ROOT***

CHORDS IN THIS KEY ·

RN	TYPE	CHORD	NOTES	FEELS LIKE…
I	Major			
ii	Minor			
iii	Minor			
IV	Major			
V	Major			
vi	Minor			
vii°	Dim.			

Remember: Roman numerals describe chord function, not just chord names.

The I chord always feels like home. · The V chord always feels unresolved. · The vi chord always feels tender or melancholy.

These feelings are the same in every key — only the note names change.

The RN and Type columns are for a major key, as on my card. Working in A minor? Cross them out and use i, ii°, III, iv, v, VI, VII (check the A minor card).

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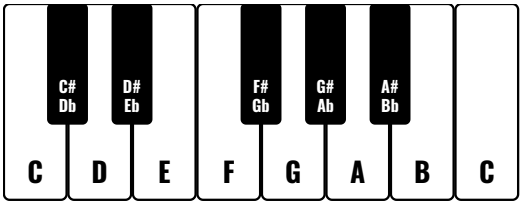
CHORDS BY EAR

The Scary Stuff · reference sheet

REFERENCE

Notes, scales, chords

Keep this open while you work. You do not need to memorize it.



The Keyboard. Every octave has the same 7 white keys and 5 black keys. The pattern repeats. Find C — it sits just to the left of every group of 2 black keys.

Half step (H): moving one key in either direction — the smallest interval. E to F and B to C are half steps on white keys. **Whole step (W):** two half steps — skip one key. C to D, D to E, F to G, G to A, A to B are whole steps.

SCALE FORMULAS ·

Major	W - W - H - W - W - W - H	*Bright, resolved, familiar*
Natural Minor	W - H - W - W - H - W - W	*Darker, more emotional than major*
Major Pentatonic	W - W - WH - W - WH	*5-note major scale — bright and simple*
Minor Pentatonic	WH - W - W - WH - W	*5-note minor scale — common in blues and rock*
Harmonic Minor	W - H - W - W - H - WH - H	*Minor with a raised 7th — dramatic, Eastern quality*
Melodic Minor	W - H - W - W - W - W - H	*Minor with raised 6th and 7th ascending*
Blues Scale	WH - W - H - H - WH - W	*Minor pentatonic + flat 5 — essential for blues and jazz*

CHORD TYPES ·

TYPE	NOTES	FORMULA	EXAMPLE (ROOT C)	SOUNDS LIKE
Major	3 notes	R - 4 - 3	C major = C - E - G	*Bright, stable, resolved*
Minor	3 notes	R - 3 - 4	C minor = C - Eb - G	*Darker, more emotional*
Diminished	3 notes	R - 3 - 3	C dim = C - Eb - Gb	*Unstable, tense, unresolved*
Augmented	3 notes	R - 4 - 4	C aug = C - E - G#	*Dreamy, unsettled*
Dominant 7th	4 notes	R - 4 - 3 - 3	C7 = C - E - G - Bb	*Bluesy, wants to resolve*
Major 7th	4 notes	R - 4 - 3 - 4	Cmaj7 = C - E - G - B	*Lush, jazzy, sophisticated*
Minor 7th	4 notes	R - 3 - 4 - 3	Cmin7 = C - Eb - G - Bb	*Smooth, melancholy, jazzy*

CHORDS IN A KEY — ROMAN NUMERALS ·

Every key has 7 chords built from its scale. The Roman numeral tells you which scale degree the chord is built on. Uppercase = major chord. Lowercase = minor chord.

I	Major	*Home. Complete. The resting point.*
ii	Minor	*Gentle tension. Moving somewhere.*
iii	Minor	*Wistful. Transitional.*
IV	Major	*Lifting. Hopeful. Away from home.*
V	Major	*Homesick. Wants to resolve back to I.*
vi	Minor	*Tender. Melancholy. Emotional center.*
vii°	Diminished	*Unstable. Tense. Rarely used alone.*

In a minor key, the i chord is home — but it feels more unsettled and emotional than the major I. The iv, v, VI, and VII chords take on the same functions relative to their new home.

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CHORDS BY EAR

Project sheet · overview, requirements, chord function

STUDENT

Name _____ Date _____

CHORDS & ROMAN NUMERALS

The Building Blocks · Part 3 · Digital Music Production 1 · Unit 3 · 3 project days (Days 3–5)

OVERVIEW ·

You have learned the keyboard and the notes of your scale. Now you are going to build the chords that give a melody its harmonic context. You will use the R-4-3 and R-3-4 formulas to construct chords, learn how Roman numerals describe their function, and build a chord layer on top of the provided bassline in Logic.

REQUIREMENTS ·

- ☐ Complete the chord construction pages and Roman numeral section in the Scary Stuff packet
- ☐ Complete your Reference Card for your chosen key (C Major or A Minor)
- ☐ Download the provided Logic file — a bassline is already written
- ☐ Build the 4 chords from the progression in Logic using the piano roll — one chord per measure
- ☐ Each chord correctly constructed using R-4-3 (major) or R-3-4 (minor)
- ☐ Label each chord region in Logic with its Roman numeral
- ☐ Identify the Roman numeral and function of each chord in your progression in writing (see below)
- ☐ Project named: Your Name — Building Blocks Part 3
- ☐ Exported as MP3 and .zip (compressed .logicx file) and submitted to your class page before presenting

CHORD FUNCTION — FILL THIS IN ·

After building your chords, complete this table:

CHORD #	ROMAN NUMERAL	CHORD NAME	WHAT IT FEELS LIKE (IN YOUR OWN WORDS)
1			
2			
3			
4			

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CHORDS BY EAR

Project sheet · tips and formulas

STUDENT

Name _____ Date _____

TIPS FOR SUCCESS ·

- Build one chord at a time and play it before moving on. If it sounds wrong, check the formula — count half steps carefully from the root.
- In the piano roll, all three notes of a chord should start at exactly the same position and last for the same duration.
- Label each region with its Roman numeral as you go. You will need these labels when you present your project.
- Your Reference Card is your most important tool here. Keep it open.

THE CHORD FORMULAS ·

MAJOR
R - 4 - 3
bright and stable

MINOR
R - 3 - 4
darker and more emotional

DIMINISHED
R - 3 - 3
tense and unstable

COUNTING HALF STEPS ·

Do not count the root. For C major (R-4-3): C#(1), D(2), D#(3), **E**(4). Then count 3 from E: F(1), F#(2), **G**(3). C major = C - E - G.

THE TEST
Point to your I chord. Play it. Does it feel like home?

NOTES ·

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CHORDS BY EAR

Key vocabulary

STUDENT

12 words

Name _____ Date _____

Read each word and its meaning. These are the words you should own by the end of the unit.

WORD	MEANING
Half step	moving one key in either direction, including black keys. The smallest possible interval.
Whole step	two half steps: skip one key. Most white key pairs are whole steps.
Scale	an ordered set of notes built with a formula of whole and half steps. The ingredient list for a key.
Key	the home a piece of music settles on: its home note, its scale and the chords built from that scale (this unit: C major or A minor).
Chord	three or more notes played at the same time, built by stacking intervals on a root.
Root	the note a chord is built on: the R in R-4-3.
Major chord	R-4-3. Sounds bright and stable.
Minor chord	R-3-4. Sounds darker and more emotional.
Diminished chord	R-3-3. Sounds tense and unstable.
Roman numeral	a chord's name from its position in the scale. Uppercase = major, lowercase = minor, ° = diminished.
Chord function	what a chord does in the key: the emotional role it plays (home, lifting, homesick, tender).
Balance	how loud each layer is compared with the others. The part the listener should follow sits on top; the rest steps back.

QUICK CHECK · ANSWERS ON PAGE 32

1. Three or more notes played at the same time are a _____. _____

2. C to C# is one _____. _____

3. The I, IV and V in I–IV–V–I are _____s. _____

4. The note a chord is built on is its _____. _____

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CHORDS BY EAR

Submission checklist

STUDENT

Before you present

Name _____ Date _____

Tick each box before you submit and present. If a box isn't ticked, you aren't done yet.

- ☐ Chord construction pages and Roman numeral section of the Scary Stuff packet completed
- ☐ Reference Card completed for your chosen key
- ☐ 4 chords built in Logic using correct formulas
- ☐ Each chord region labeled with its Roman numeral
- ☐ Chord function table filled in above
- ☐ Project named: Your Name — Building Blocks Part 3
- ☐ Exported as MP3 and .zip and submitted to your class page before presenting

GET READY TO PRESENT ·

My four chords (Roman numerals): _____

My home chord is _____ **because** _____

The chord that wants to go home is _____

NOTES ·

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CHORDS BY EAR

Rubric

RUBRIC

/ 15

Name _____ Date _____

PART 3: CHORDS & ROMAN NUMERALS — RUBRIC

The Building Blocks · Unit 3 · /15

Each category is scored from 2 to 5.

Category	2	3	4	5	Score
CHORD CONSTRUCTION	Chords built with wrong notes or polyphonic errors. Formula not applied correctly.	Most chords correct but at least one has a construction error. Some formula misapplication.	All chords correctly constructed using R-4-3 or R-3-4. Notes stacked correctly in piano roll.	All chords correct, well-voiced, and labeled with Roman numerals. Demonstrates understanding of the formula.	
HARMONIC UNDERSTANDING	Chord function table incomplete or descriptions show no understanding of how chords feel.	Table complete but descriptions are vague or generic (e.g. 'sounds good').	Table complete. Student can describe what each chord feels like in their own words.	Table complete with specific, accurate descriptions. Student clearly understands why each chord was chosen.	
DOCUMENTATION & PRESENTATION	Not submitted or not presented.	Submitted late or named incorrectly. Minimal presentation engagement.	Correctly named and submitted. Presentation made. Reference Card completed.	Correctly named, on time. Student explains chord choices using Roman numeral language.	
				TOTAL	**/ 15**

MY I CHORD FEELS LIKE HOME BECAUSE...

_____

_____

_____

TEACHER COMMENTS

_____

_____

_____

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CHORDS BY EAR

Answer key · packet Parts 1–3

ANSWER KEY

Teacher

PART 1 · KEYBOARD AND STEPS

White keys: C D E F G A B. B to C **H** · E to F **H** · D to E **W** · A to B **W** · F to G **W** · G# to A **H**.

PART 2 · SCALES

C major: C D E F G A B C
A natural minor: A B C D E F G A
C major pentatonic: C D E G A C
Missing from the pentatonic: **F and B**

Part 3 · chords (every spelling counted from the formula; D major and B minor use a black key, D diminished uses A-flat, written Ab):

MAJOR CHORDS · R-4-3 — SOUNDS BRIGHT AND STABLE ·

C MAJOR

C	E	G
Root	3rd	5th

F MAJOR

F	A	C
Root	3rd	5th

G MAJOR

G	B	D
Root	3rd	5th

D MAJOR

D	F#	A
Root	3rd	5th

MINOR CHORDS · R-3-4 — SOUNDS DARKER AND MORE EMOTIONAL ·

A MINOR

A	C	E
Root	3rd	5th

D MINOR

D	F	A
Root	3rd	5th

E MINOR

E	G	B
Root	3rd	5th

B MINOR

B	D	F#
Root	3rd	5th

DIMINISHED CHORDS · R-3-3 — SOUNDS TENSE AND UNSTABLE ·

B DIMINISHED

B	D	F
Root	3rd	5th

D DIMINISHED

D	F	Ab
Root	3rd	5th

VOCABULARY · QUICK CHECK

1. chord · **2.** half step · **3.** Roman numeral · **4.** root

ROMAN NUMERALS · HEAR IT

I–IV–V–I in C: C, F, G, C. i–VI–III–VII in A minor: Am, F, C, G. The pull home: V (G major) in C; in A minor, VII (G major) drives back toward i.

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CHORDS BY EAR

Answer key · KCSRN and the Feel Log

ANSWER KEY

Teacher

MAJOR KEY

Major Key: C major

Scale:

C D E F G A B C

CHORD NAMES

I	C Major
ii	D Minor
iii	E Minor
IV	F Major
V	G Major
vi	A Minor
vii°	B Dim.

CHORD SPELLINGS

I:	C	E	G
ii:	D	F	A
iii:	E	G	B
IV:	F	A	C
V:	G	B	D
vi:	A	C	E
vii°:	B	D	F

MINOR KEY

Minor Key: A minor

Scale (natural minor):

A B C D E F G A

CHORD NAMES

i	A Minor
ii°	B Dim.
III	C Major
iv	D Minor
v	E Minor
VI	F Major
VII	G Major

CHORD SPELLINGS

i:	A	C	E
ii°:	B	D	F
III:	C	E	G
iv:	D	F	A
v:	E	G	B
VI:	F	A	C
VII:	G	B	D

CHORD FEEL LOG · WORDS FROM MY CARDS (ANY HONEST DESCRIPTION COUNTS) ·

CHORD	WHAT IT IS	FEELS LIKE (MY CARD)	USUALLY
1 · C–E–G	I in C · C Major	*Home. Settled. Complete.*	bright
2 · F–A–C	IV in C · F Major	*Lifting. Hopeful. Away from home.*	bright
3 · G–B–D	V in C · G Major	*Homesick. Wants to resolve to I.*	bright
4 · A–C–E	vi in C · A Minor	*Tender. Melancholy. Emotional.*	dark
5 · D–F–A	ii in C · D Minor	*Gentle tension. Moving somewhere.*	dark
6 · E–G–B	iii in C · E Minor	*Wistful. In between.*	dark
7 · B–D–F	vii° in C · B Dim.	*Tense. Unstable. Rarely used alone.*	tense

Loop 1 is I–IV–V–I in C major and comes home on C–E–G. Loop 2 is i–VI–III–VII in A minor: home is A–C–E. The diminished chord (B–D–F) is neither bright nor dark: accept either circle, or “tense”.

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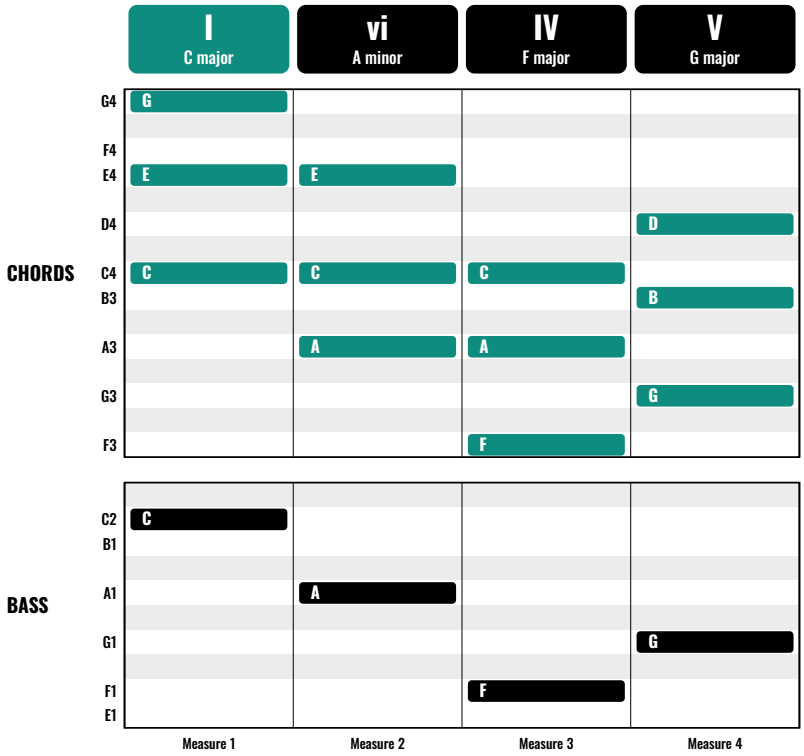
CHORDS BY EAR

Example notes · a worked example in both keys

EXAMPLE

Teacher

One possible answer for the Part 3 Logic file: four chords, one per measure, root position, over a bassline that plays each chord's root. C major file: **I–vi–IV–V**; A minor file: **i–VI–III–VII**. Teal regions: the chords students build; black: the provided bassline.



HOW TO READ IT

- Each column is one measure: one chord, one bass note.
- The three teal notes are stacked: same start, same length. Root on the bottom.
- The black bass note is the chord's root, an octave or two lower.
- Label each region with the Roman numeral on top: I, vi, IV, V.
- Grey rows are black keys. Count rows to check a chord: 4 + 3 is major, 3 + 4 is minor.

CHORD FUNCTION TABLE · C MAJOR FILE ·

CHORD #	ROMAN NUMERAL	CHORD NAME	BASS	WHAT IT FEELS LIKE (FROM THE CARD)
1	I	C Major (C - E - G)	C	*Home. Settled. Complete.*
2	vi	A Minor (A - C - E)	A	*Tender. Melancholy. Emotional.*
3	IV	F Major (F - A - C)	F	*Lifting. Hopeful. Away from home.*
4	V	G Major (G - B - D)	G	*Homesick. Wants to resolve to I.*

CHORD FUNCTION TABLE · A MINOR FILE ·

CHORD #	ROMAN NUMERAL	CHORD NAME	BASS	WHAT IT FEELS LIKE (FROM THE CARD)
1	i	A Minor (A - C - E)	A	*Home. But darker and more unsettled than major.*
2	VI	F Major (F - A - C)	F	*Warm and open. Common resolution point.*
3	III	C Major (C - E - G)	C	*Relative major. Bright moment in a darker key.*
4	VII	G Major (G - B - D)	G	*Lifting. Drives back toward i strongly.*

Same four chords, two homes: A minor is the relative minor of C major. A student who writes only "sounds good" scores a 3 for Harmonic Understanding; "wants to go back to the I" is the kind of answer the 5 describes.

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